

Energy Conversion System

Fill in the Blanks — Complete Exam Preparation

■ Coal & Types

1. In a coal-based power plant, oil is used as _____ support during short-time emergency.

Ans: *Short-time*

2. COAL is formed underground with vegetable matter over millions of years under _____ and _____.

Ans: *pressure, temperature*

3. Five types of coal in order of maturity: Peat Coal (worst), Lignite, Sub-Bituminous, Bituminous, and _____.

Ans: *Anthracite*

4. Peat is the _____ category with maximum moisture content, not considered for power generation.

Ans: *lowest*

5. Anthracite coals are most mature with maximum _____ content and highest calorific value per kg.

Ans: *carbon*

6. Indian coals are mostly _____ or _____ with high ash but low sulphur.

Ans: *Sub-bituminous, Bituminous*

■ Coal Analysis

7. Ultimate Analysis shows the _____ composition of coal in addition to moisture and ash.

Ans: *elemental*

8. The major elemental composition of coal is represented as _____.

Ans: *CHONS (Carbon, Hydrogen, Oxygen, Nitrogen, Sulphur)*

9. Moisture can be of two types: _____ Moisture and _____ Moisture.

Ans: *Surface, Inherent*

10. Surface moisture is evaporated _____ with any sort of heating.

Ans: *immediately*

11. Inherent moisture is within the mass of coal and difficult to remove without _____ processes.

Ans: *combustion*

12. Proximate Analysis originates following actual combustion of a coal particle in a heated surrounding similar to a _____.

Ans: *Boiler furnace*

13. The released gas from heating coal is called _____.

Ans: Volatile Matter

14. The solid carbon called _____ and incombustible material (metal oxides) called _____ remain after combustion.

Ans: Fixed Carbon, Ash

15. Proximate analysis is done as: _____, _____, _____, _____.

Ans: VM (Volatile Matter), FC (Fixed Carbon), M (Moisture), Ash

■ Air Requirement & Combustion

16. Stoichiometric air of a fuel is the required air for _____ reaction during combustion.

Ans: chemical

17. Stoichiometric air is the _____ possible air required for theoretically complete combustion.

Ans: minimum

18. In power plants, pulverized coal is used for possible _____ mixing with air.

Ans: better

19. Excess air contributes to _____ gas loss as it carries heat away from the fuel to the chimney.

Ans: Exhaust

20. The intersection point of decreasing incomplete combustion loss and increasing exhaust gas loss is called the _____ of the A/F ratio.

Ans: Operating Zone

■ Calorific Value (HHV & LHV)

21. The heat release including water droplet condensation is called the _____.

Ans: HHV — Higher Heating Value

22. If the final exit of flue gas is without condensation of water vapour, the corresponding lower value is known as _____.

Ans: LHV — Lower Heating Value

23. $LHV = HHV -$ _____.

Ans: $m_w \times h_{fg}$

24. The dew point temperature is also called _____.

Ans: Acid-Condensation Temperature

25. In power plants, the exit temperature of flue gas cannot be allowed to be below _____.

Ans: Acid-Condensation Temperature

■ Boiler Losses & Efficiency

26. The Direct method of Boiler Efficiency measurement gives the ratio of _____ output and _____ input.

Ans: *desired, required*

27. The Direct method does not help to identify possible measures to _____ the efficiency of the boiler.

Ans: *improve*

28. L1 represents _____ flue gas loss.

Ans: *Dry*

29. L2 represents heat loss due to _____.

Ans: *Moisture*

30. L3 represents loss due to _____ in fuel.

Ans: *Hydrogen*

31. L4(a) represents _____ carbon found in ash.

Ans: *Unburned*

32. L4(b) represents _____ formation instead of CO₂ formation.

Ans: *CO*

33. $L_5 = L_{\text{total}} - (L_1 + L_2 + L_3 + L_4)$ is called _____ loss.

Ans: *Unaccounted*

■ Pulverized Coal & PF Boiler

34. The coal-carrying air in PF boilers is called _____.

Ans: *Primary Air*

35. Oxygen of Primary Air supports _____ combustion only, not complete combustion.

Ans: *partial*

36. Additional air supplied at different stages of combustion is called _____, _____ air, etc.

Ans: *Secondary, Tertiary*

37. Force Draft Fan supplies both _____ and _____.

Ans: *PA (Primary Air), SA (Secondary Air)*

38. Fly-Ash generated from pulverized coal can be reused in the form of making _____ contributing to _____ concept.

Ans: *Bricks, Waste-Wealth*

39. Power plant operators use the _____ process using steam from the boiler to clean the outside heat transfer surface.

Ans: *Soot Blowing*

■ Burner Arrangements

40. Three types of burner arrangements are: _____, _____, and _____.

Ans: *Front firing, Opposed firing, Corner/Tangential firing*

41. In corner-fired furnaces, A/F mixture enters _____ to a constricted swirl and combustion looks like a _____.

Ans: *tangentially, Fire Ball*

■ Boiler Circulation & Feed Water Heating

42. Feed water heating in a boiler happens in four stages: _____, Evaporation, Superheater, and _____.

Ans: *Economiser, Reheater*

43. Tubes coming down from the bottom of the boiler drum are called _____.

Ans: *Downcomers*

44. Tubes from the header connected back to the drum are called _____.

Ans: *Risers*

45. The mechanism of water flow through risers and downcomers including the drum is called _____.

Ans: *Circulation*

46. Circulation may be _____ (using a pump) or _____ (due to density difference).

Ans: *Forced, Natural*

47. Superheaters receiving heat maximum in radiation mode are called _____.

Ans: *Radiant Superheaters*

48. Any Superheater/Reheater located in the second pass is called _____ Superheater/Reheater.

Ans: *Convective*

■ Steam Turbine — Advantages

49. Out of different prime movers, _____ have emerged as the best possible option for large scale power generation.

Ans: *Steam Turbines*

50. Steam turbines with more than 1000 MW of about _____ efficiency are available in the market.

Ans: *60%*

51. Steam turbine generates power in a _____ motion, requiring no conversion from reciprocating motion.

Ans: *rotating*

52. Steam turbine delivers continuous constant power against a constant resisting torque — called _____ advantage.

Ans: *Operational*

53. In turbines, steam does not get _____ with the lubricating oil as no lubricating oil is used in the steam space.

Ans: contaminated

■ Classification of Steam Turbines

54. On the basis of steam input, turbines may be _____ pressure or _____ pressure.

Ans: Single, Dual/Multi

55. If the whole amount of steam from turbine goes to condenser, it is called _____ Turbine.

Ans: Condensing

56. If whole steam passes through a process heater for utility heating, it is called _____ Turbine.

Ans: Back Pressure

57. If a small part of steam is extracted for process heating and the rest passes through condenser, it is called _____ Turbine.

Ans: Condensing Extraction Turbine (CET)

■ Impulse & Reaction Turbines

58. In Impulse Turbine, the enthalpy of steam is fully converted to _____ steam jet through nozzles before blades.

Ans: high velocity

59. For Simple Impulse Turbine, the exit velocity is very high — about _____ of inlet velocity.

Ans: 33%

60. In Pressure Compounding (Rateau Turbine), total pressure drop of steam occurs in successive rows of _____.

Ans: nozzles

61. In Velocity Compounding (Curtis Turbine), the whole pressure drop occurs in a _____ row of nozzle.

Ans: single

62. The function of Fixed Blades in a Curtis Turbine is only to change the _____ of the steam coming out from MB1.

Ans: direction

63. In Reaction Turbine, power is generated due to _____ force out of pressure drop inside the moving part.

Ans: reaction

64. Pure Reaction Turbine (Heron's Turbine) — high pressure steam enters _____ and expands through two nozzles radially.

Ans: centrally

65. Impulse Reaction Turbines used in power plants are known as _____ or 50% Reaction Turbine.

Ans: *Parsons' Turbine*

66. In Impulse Reaction Turbine, pressure drop occurs both in _____ (Stator) and _____ (Rotor).

Ans: *nozzles, moving blades*

67. Optimal speed ratio for maximum blade efficiency: $p_{opt} =$ _____.

Ans: $\cos(\alpha_1) / 2$

68. Maximum Blade Efficiency (for $k, c = 1$): $\eta_{B,max} =$ _____.

Ans: $\cos^2(\alpha_1)$

69. Degree of Reaction for identical stator and rotor blades = _____, and this turbine is called _____.

Ans: *50%, Parsons' Turbine*

■ Varying Cross Section Flow (Nozzles)

70. Flow of a compressible fluid through a passage with varying cross-sectional area is called _____ flow.

Ans: *Varying Cross Section / Isentropic*

71. For subsonic velocity at inlet ($M < 1$), a diverging passage will behave as a _____.

Ans: *Diffuser*

72. For supersonic flow ($M > 1$), a divergent flow path will behave as a _____.

Ans: *Nozzle*

73. To convert steam velocity from very low (subsonic) to supersonic, a _____ nozzle is used.

Ans: *Convergent-Divergent (C-D)*

74. At the throat of a properly designed C-D nozzle, the velocity will be _____.

Ans: *Sonic*